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component name; the component definition using programming-specific terminology; and detailed descriptions of the code component's meaning and explicit hint to its placement (Fig. 1).

Active learning is also supported by an AR workspace (Figs. 2, 3) with interactive visualisation assets (HMD-anchored glossary; localised virtual code template, program visualisation (PV), and progress indicator) that serve as additional guides for learners for constructing their program. Physical manipulatives are placed into the template slots, and a simplified parsing mechanism activated by a virtual button checks the statement and displays the result as a textual and visual cue. Immediate visual feedback is provided by colour-coding individual slots based on the state of the construction.

A static PV depicts the structure and meaning of the programming statement in a non-sequential form (Fig. 3). Static visualisations allow interaction with knowledge embodied inside the PV at learners' own pace and with minimal cognitive load [12]. Non-sequential deconstruction of the statement is designed to stimulate curiosity and solidify it by including an interactive, proximity-based annotation system. An interactive node-based "astronomy" metaphor is used for contextualising instantiation concepts, with the "planets" denoting primary concepts and "satellites" that orbit the planets indicating secondary concepts (expressions, operators, etc.) that bridge the primary concepts.

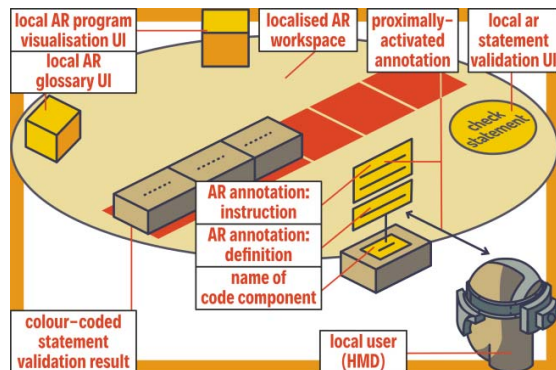


Figure 2: Schematic of the system's TUI with interactive AR tooltips and validation engaged in a statement construction scenario.

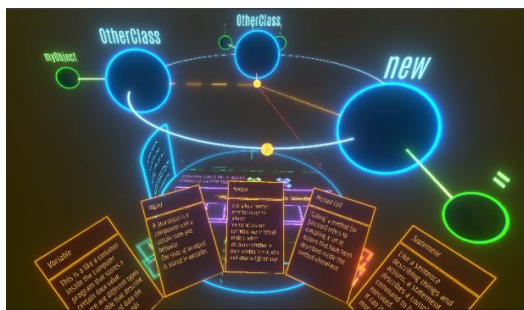


Figure 3: Localised AR components: contextual user-anchored glossary of programming terms; and static PV.

Additional information on relevant programming terminology used in AR tooltips for code blocks is provided as text cues in a gesture-toggled glossary (Fig. 3), provides contextual knowledge to learners.

3 CONCLUSION AND FUTURE WORK

This project contributes to the development of PL environments by suggesting how TUI and AR (HMD) may support a CSE-specific model of PL [1, 2] (as a two-dimensional activity), and

implement advanced pedagogical theories and principles (active guided learning; embodied learning; spatial learning; scaffolding; gamification; feedback) while considering the cognitive load aspect. The suggested system design situates and contextualises OOP concepts within the learners' physical environment through HMD-supported AR and tangible manipulatives. Key components of the system design include: block-anchored AR annotation; localised AR program visualisation; HMD-anchored AR glossary; and AR- and TUI-based interaction. The two-dimensional structure of the PL environment and the need to support active guided learning by embracing affordances of AR and TUIs determined the choice of a HMD (at this stage) as a method that best corresponds to the complexity of the learning content and activity, and a medium that may decrease cognitive load by enabling natural interaction with learning materials. Future work will involve a field study for comparative evaluation of HMD and HMD prototypes in terms of user experience, performance, cognitive load, and engagement.

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